

Theme 5 Definitions

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5.1 Antiderivatives

Definitio A function F is called an **antiderivative** of f on an interval I if $F'(x) = f(x)$ for all x in I .

1 Theorem If F is an antiderivative of f on an interval I , then the most general antiderivative of f on I is

$$F(x) + C$$

where C is an arbitrary constant.

2 Table of Antidifferentiation Formulas

Function	Particular antiderivative	Function	Particular antiderivative
$cf(x)$	$cF(x)$	$\sin x$	$-\cos x$
$f(x) + g(x)$	$F(x) + G(x)$	$\sec^2 x$	$\tan x$
$x^n \ (n \neq -1)$	$\frac{x^{n+1}}{n+1}$	$\sec x \tan x$	$\sec x$
$\frac{1}{x}$	$\ln x $	$\frac{1}{\sqrt{1-x^2}}$	$\sin^{-1} x$
e^x	e^x	$\frac{1}{1+x^2}$	$\tan^{-1} x$
b^x	$\frac{b^x}{\ln b}$	$\cosh x$	$\sinh x$
$\cos x$	$\sin x$	$\sinh x$	$\cosh x$

5.3 The Definite Integral

2 Definition of a Definite Integral If f is a function defined for $a \leq x \leq b$, we divide the interval $[a, b]$ into n subintervals of equal width $\Delta x = (b - a)/n$. We let $x_0 (= a), x_1, x_2, \dots, x_n (= b)$ be the endpoints of these subintervals and we let $x_1^*, x_2^*, \dots, x_n^*$ be any **sample points** in these subintervals, so x_i^* lies in the i th subinterval $[x_{i-1}, x_i]$. Then the **definite integral of f from a to b** is

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

provided that this limit exists and gives the same value for all possible choices of sample points. If it does exist, we say that f is **integrable** on $[a, b]$.

3 Theorem If f is continuous on $[a, b]$, or if f has only a finite number of jump discontinuities, then f is integrable on $[a, b]$; that is, the definite integral $\int_a^b f(x) dx$ exists.

4 Theorem If f is integrable on $[a, b]$, then

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x$$

where $\Delta x = \frac{b - a}{n}$ and $x_i = a + i \Delta x$

Comparison Properties of the Integral

6. If $f(x) \geq 0$ for $a \leq x \leq b$, then $\int_a^b f(x) dx \geq 0$.

7. If $f(x) \geq g(x)$ for $a \leq x \leq b$, then $\int_a^b f(x) dx \geq \int_a^b g(x) dx$.

8. If $m \leq f(x) \leq M$ for $a \leq x \leq b$, then

$$m(b - a) \leq \int_a^b f(x) dx \leq M(b - a)$$

If f is integrable on an interval $[a, b]$ and $f(x) \geq 0$ for every $x \in [a, b]$, then $\int_a^b f(x) dx \geq 0$. In fact, $\int_a^b f(x) dx$ is equal to the area of the region between the graph of f , the x -axis and the lines $x = a$ and $x = b$.

If f is integrable on an interval $[a, b]$ and $f(x) \leq 0$ for every $x \in [a, b]$, then $\int_a^b f(x) dx \leq 0$. In fact, $\int_a^b f(x) dx$ is equal to the negative of the area of the region between the graph of f , the x -axis and the lines $x = a$ and $x = b$.

5.4 The Indefinite Integral

$$\int f(x) dx = F(x) \quad \text{means} \quad F'(x) = f(x)$$

For example, we can write

$$\int x^2 dx = \frac{x^3}{3} + C \quad \text{because} \quad \frac{d}{dx} \left(\frac{x^3}{3} + C \right) = x^2$$

So we can regard an indefinite integral as representing an entire *family* of functions (one antiderivative for each value of the constant C).